MINISTERUL EDUCAȚIEI ȘI CERCETĂRII

UNIVERSITATEA TEHNICĂ
DIN CLUJ-NAPOCA

HYDRAULIC AND PNEUMATIC AUTOMATION EQUIPMENT PROJECT			
Student Name	Motoc Amelie-Iulia	GROUP:	30132

Comparative Study of Linear vs. Non-linear Control for a Pneumatic Artificial Muscle System

ACADEMIC YEAR: 2025-2026

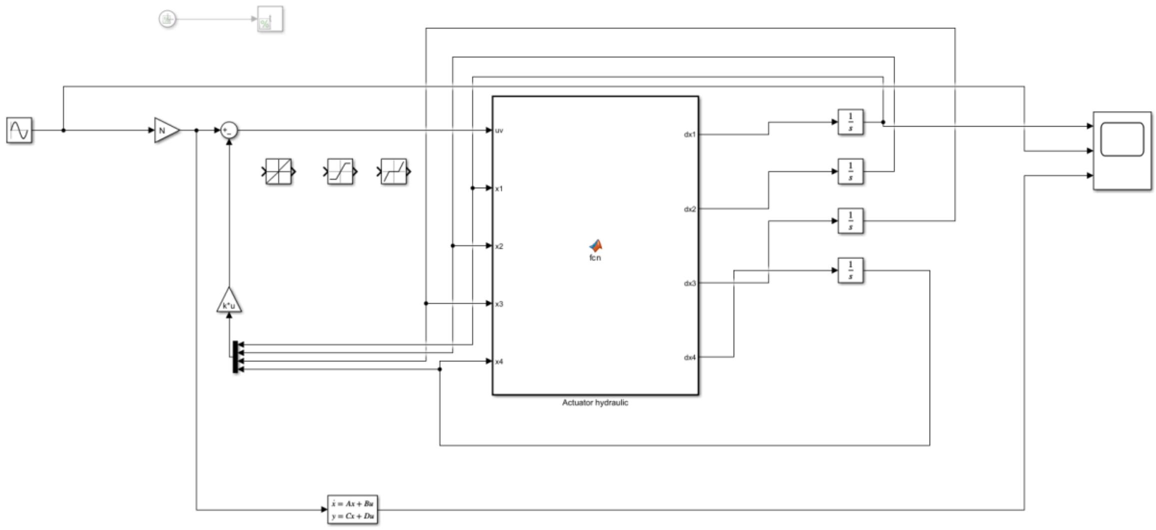

Figura1: „ Simulink Diagram"

The Simulink model implements a closed-loop control architecture. It evaluates the performance of the idealized linearized system side-by-side and in parallel with the real non-linear dynamics of the pneumatic artificial muscle

1. Low Frequency Operation (0.5 Hz): Both the linearized and non-linear models track the sinusoidal reference signal with exceptional accuracy. System attenuation is minimal, and the dynamic phase lag remains virtually unnoticeable.

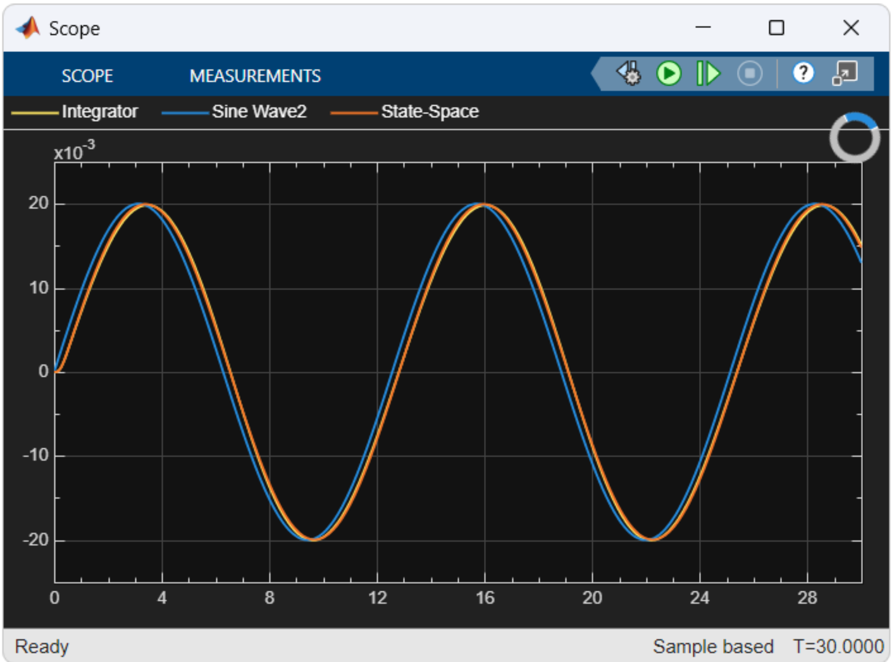

Figure 2: Closed-Loop Dynamic Response at Low Frequency

2. Medium Frequency Operation (1.0 Hz): While both mathematical models successfully maintain trajectory tracking, a visible phase lag begins to manifest across the timeline.

A minor amplitude degradation (attenuation) becomes observable at the peak wave crests and troughs of the response.

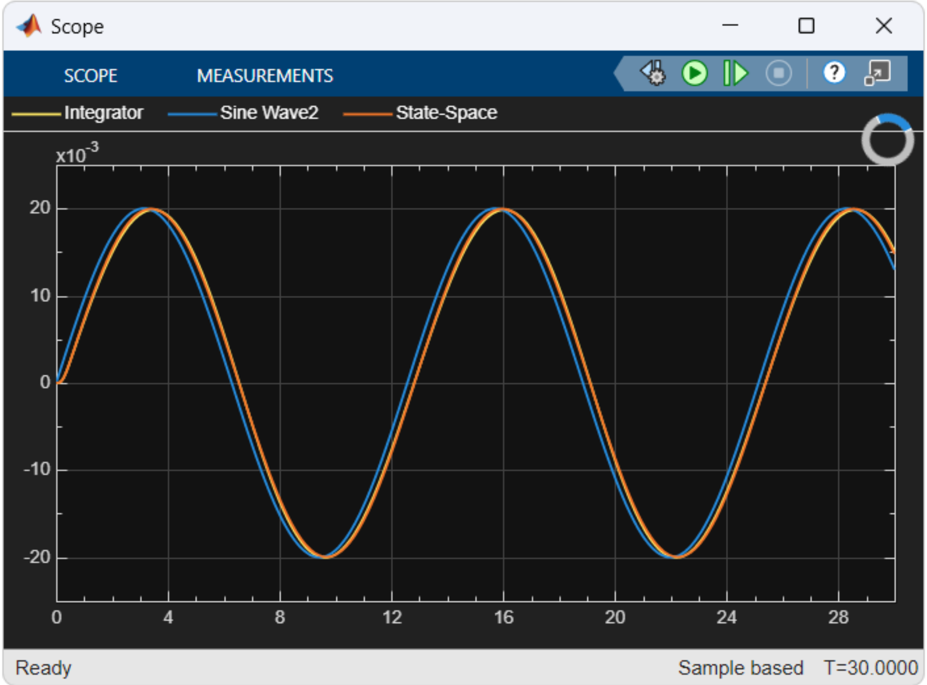

Figure 3: Closed-Loop Dynamic Response at Medium Frequency

3. High Frequency Operation (10.0 Hz) : The reference excitation changes much too rapidly for the inherent low-pass structural dynamics of pneumatic systems to cope. This behavior explicitly demonstrates that under high-frequency transient states, localized non-linearities scale up aggressively, causing the linearized tangent model derived in MATLAB to lose its localized operating-point precision.

-

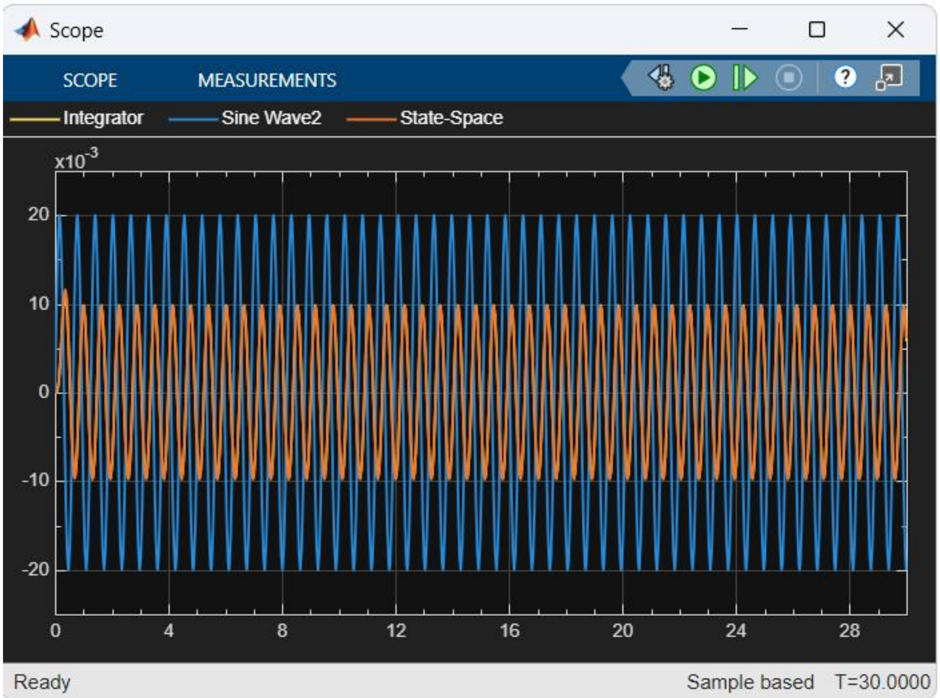

Figure 4: Closed-Loop Dynamic Response at High Frequency